

Task 11

Description:

Deploy a simple web application on AWS that is accessible through a load balancer and sends an email alert when the server's CPU utilization is high.

Services to Use

- IAM
- EC2
- EBS
- Application Load Balancer (ALB)
- CloudWatch
- SNS

Task Requirements

- Launch an **EC2 instance** with an **IAM role** attached (no access keys).
- Attach and mount an **EBS volume** to the EC2 instance.
- Install Apache or Nginx and host a sample web page.
- Place the EC2 instance behind an **Application Load Balancer**.
- Enable **CloudWatch monitoring** for the EC2 instance.
- Create a **CloudWatch alarm** for high CPU utilization.
- Configure **SNS** to send an email notification when the alarm is triggered.

Expected Outcome

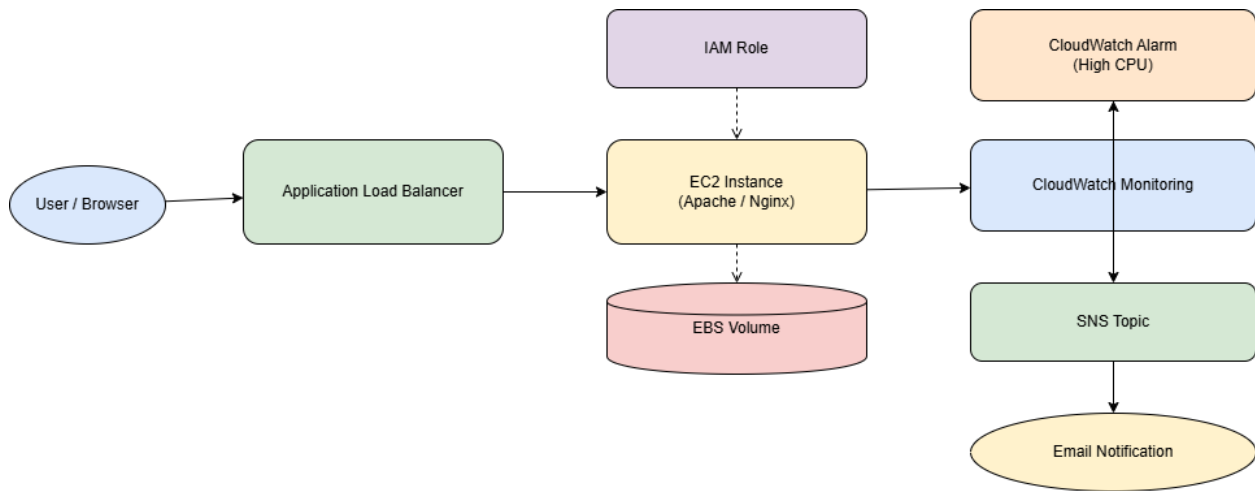
- Web application accessible via ALB DNS name.
- EC2 instance using IAM role securely.
- Email notification received when CPU threshold is crossed.

Deliverables

- Screenshots of:
 - EC2 instance and EBS attachment
 - ALB URL showing web page
 - CloudWatch alarm
 - SNS email alert
- Brief explanation of how the services are integrated.

Solution:

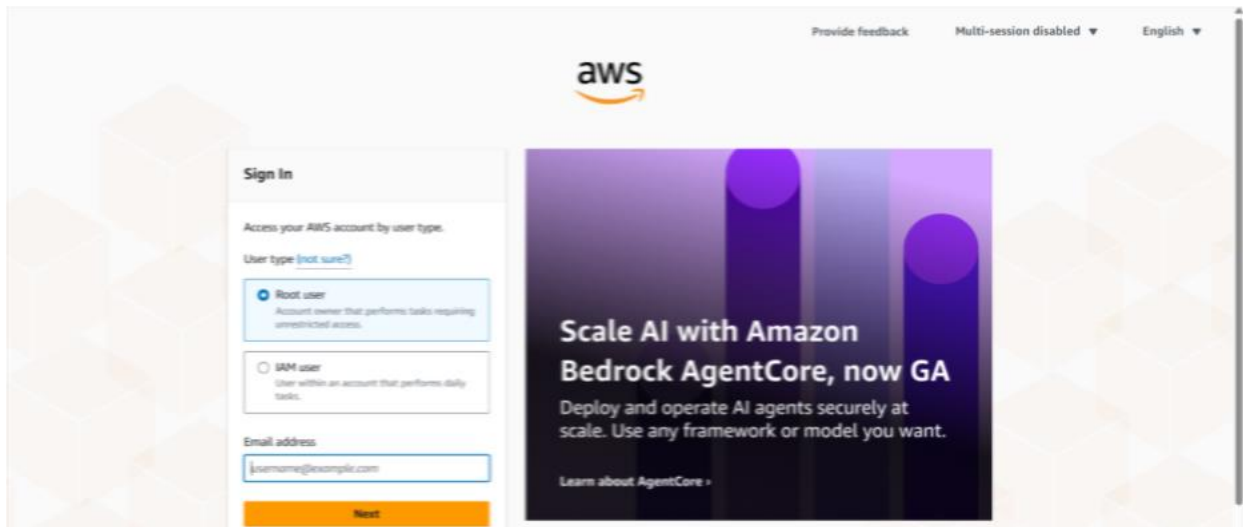
Workflow -



1. Login to your AWS account with credentials and required verification.

(<https://signin.aws.amazon.com/>)

My account – AWS Management Console- Sign in



2. Create IAM Role for EC2 (No Access Keys)
 - Open AWS Console -IAM – Create Roles
 - Trusted entity:
 - Select AWS service
 - Use case: EC2
 - Attach Permissions

- *AmazonEC2ReadOnlyAccess*
- *CloudWatchAgentServerPolicy*
- *AmazonSNSFullAccess*
- Create Role

The screenshot shows the 'Create role' page in the AWS IAM console. The left sidebar indicates the current step is 'Step 3: Name, review, and create'. The main content area is divided into two sections:

- Trusted entity type:** This section contains five radio button options:
 - AWS service** (selected): Allow AWS services like EC2, Lambda, or others to perform actions in this account.
 - AWS account:** Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.
 - Web identity:** Allow users federated by the specified external web identity provider to assume this role to perform actions in this account.
 - SAML 2.0 federation:** Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.
 - Custom trust policy:** Create a custom trust policy to enable others to perform actions in this account.
- Use case:** This section is for allowing an AWS service like EC2, Lambda, or others to perform actions in this account. It includes a dropdown menu for 'Service or use case' with 'EC2' selected. Below this, it says 'Choose a use case for the specified service.' and provides two options:
 - EC2** (selected): Allows EC2 instances to call AWS services on your behalf.
 - EC2 Role for AWS Systems Manager:** Allows EC2 instances to call AWS services like CloudWatch and Systems Manager on your behalf.

The screenshot shows the 'Create role' page in the AWS IAM console, specifically the 'Step 2: Add permissions' and 'Step 3: Add tags' sections. The left sidebar indicates the current step is 'Step 2: Add permissions'.

Step 2: Add permissions

The 'Permissions policy summary' table shows the following policies attached to the role:

Policy name	Type	Attached as
AmazonEC2ReadOnlyAccess	AWS managed	Permissions policy
AmazonSNSFullAccess	AWS managed	Permissions policy
CloudWatchAgentServerPolicy	AWS managed	Permissions policy

Step 3: Add tags

The 'Add tags - optional' section shows that there are no tags associated with the resource. It includes an 'Add new tag' button and a note that you can add up to 50 more tags.

3. Launch EC2 Instance with IAM Role

- EC2 - Instances - Launch Instance
- Name: WebServer
- AMI: Amazon Linux 2
- Instance type: t2.micro
- Key pair: Select or create one
- Attach IAM Role
- Security Group
 - Allow:
 - HTTP → Port 80 → Anywhere

- SSH → Port 22 → Your IP
- Launch Instance

EC2 > Instances > Launch an instance

Launch an instance

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags

Name

Myweb-server

Add additional tags

Application and OS Images (Amazon Machine Image)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Search

Search our full catalog including 1000s of application and OS images

Recents

Quick Start

Amazon Linux

macOS

Ubuntu

Windows

Red Hat

SUSE Linux

Debian

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Summary

Number of instances

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.10...[read more](#)

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs. 750 hours per month of

Cancel

Launch instance

Preview code

EC2 > Instances > Launch an instance

Instance type

Instance type

t3.micro

Family: t3 2 vCPU 1 GiB Memory Current generation: true On-Demand Linux base pricing: 0.0112 USD per Hour On-Demand SUSE base pricing: 0.0112 USD per Hour On-Demand Windows base pricing: 0.0204 USD per Hour On-Demand Ubuntu Pro base pricing: 0.0147 USD per Hour On-Demand RHEL base pricing: 0.04 USD per Hour

Free tier eligible

All generations

Compare instance types

Key pair (login)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

mykey

Create new key pair

Network settings

Network

vpc-08759b5fe29f72e8d

Subnet

No preference (Default subnet in any availability zone)

Edit

Summary

Number of instances

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.10...[read more](#)

Virtual server type (instance type)

t3.micro

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Cancel

Launch instance

Preview code

EC2 > Instances > Launch an instance

Network settings

VPC - required

vpc-04f2c67356d87b37a (My-VPC--vpc)

Subnet

subnet-0276eadab9a1d009d

My-VPC--subnet-public1-ap-south-1a

Auto-assign public IP

Enable

Firewall (security groups)

Create security group

Select existing security group

Security group name - required

Myweb-server

Description - required

launch-wizard-1 created 2026-01-13T07:28:29.123Z

Summary

Number of instances

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.10...[read more](#)

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs. 750 hours per month of

Cancel

Launch instance

Preview code

EC2 > Instances > Launch an instance

Inbound Security Group Rules

Security group rule 1 (TCP, 22, 111.125.244.2/32) Remove

Type	Protocol	Port range	Name	Description - optional
ssh	TCP	22	111.125.244.2/32	e.g. SSH for admin desktop

Security group rule 2 (TCP, 80, 0.0.0.0/0) Remove

Type	Protocol	Port range	Source	Description - optional
HTTP	TCP	80	0.0.0.0/0	e.g. SSH for admin desktop

Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

[Add security group rule](#)

Summary

Number of instances: 1

Software Image (AMI): Amazon Linux 2023 AMI 2023.10...[read more](#)
ami-0ced6a024bb18ff2e

Virtual server type (instance type): t3.micro

Firewall (security group): New security group

Storage (volumes): 1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

0 x File systems Edit

Advanced details

Domain join directory: Select Create new directory

IAM instance profile: task-11-IAMRole Create new IAM profile

Hostname type: IP name

DNS Hostname: ☒ Enable IP name IPv4 (A record) DNS requests
☐ Enable resource-based IPv4 (A record) DNS requests
☐ Enable resource-based IPv6 (AAAA record) DNS requests

Instance auto-recovery: Select

Shutdown behavior: [Info](#)

Summary

Number of instances: 1

Software Image (AMI): Amazon Linux 2023 AMI 2023.10...[read more](#)
ami-0ced6a024bb18ff2e

Virtual server type (instance type): t3.micro

Firewall (security group): New security group

Storage (volumes): 1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

4. Attach and Mount EBS Volume

- EC2 - Elastic Block Store - Volumes
- Create Volume:
 - Type: gp2
 - Size: 5 GB
 - Availability Zone: same as EC2
- Attach Volume
 - Select volume - Actions - Attach Volume
 - Choose your EC2 instance

Create volume [Info](#)

Create an Amazon EBS volume to attach to any EC2 instance in the same Availability Zone.

Volume settings

Volume type [Info](#)

General Purpose SSD (gp2)

Size (GiB) [Info](#)

10

Min: 1 GiB, Max: 16384 GiB.

IOPS [Info](#)

100 / 3000

Baseline of 3 IOPS per GiB with a minimum of 100 IOPS, burstable to 3000 IOPS.

Throughput (MiB/s) [Info](#)

Not applicable

Availability Zone [Info](#)

aps1-az1 (ap-south-1a)

Snapshot ID - optional [Info](#)

Don't create volume from a snapshot

Encryption [Info](#)

Attach volume [Info](#)

Attach a volume to an instance to use it as you would a regular physical hard disk drive.

Basic details

Volume ID

vol-058ed95bfb6128f50

Availability Zone

aps1-az1 (ap-south-1a)

Instance [Info](#)

i-005688ba2cc38c349
(Myweb-server) (running)

Only instances in the same Availability Zone as the selected volume are displayed.

Device name [Info](#)

/dev/sdk

Recommended device names for Linux: /dev/xvda for root volume, /dev/sd[f-p] for data volumes.

Newer Linux kernels may rename your devices to /dev/xvdf through /dev/xvdp internally, even when the device name entered here (and shown in the details) is /dev/sdf through /dev/sdp.

[Cancel](#) [Attach volume](#)

EC2

Dashboard
AWS Global View
Events

Instances

Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager

Images

AMIs
AMI Catalog

Successfully attached volume vol-058ed95bfb6128f50 to instance i-005688ba2cc38c349.

Volumes (2) [Info](#)

Choose filter set

	Name	Volume ID	Type	Size	IOPS	Throughput	Snapshot ID	Source volume ID	Created
☐		vol-0644a46bc6932f16	gp3	8 GiB	3000	125	snap-07e5239...	-	2026/01/13 13:05 GMT+!
☐		vol-058ed95bfb6128f50	gp2	10 GiB	100	-	-	-	2026/01/13 13:10 GMT+!

Fault tolerance for all volumes in this Region

Snapshot summary

Recently backed up volumes / Total # volumes

0 / 1

Data Lifecycle Manager default policy for EBS Snapshots status

No default policy set up | [Create policy](#)

- Mount EBS on EC2

```

10 sudo mkfs -t ext4 /dev/nvme1n1
11 lsblk
12 sudo mkfs -t ext4 /dev/nvme1n1
13 sudo mount /dev/nvme1n1 /data
14 df -h
15 histoty

```

```

[ec2-user@ip-10-0-11-59 ~]$ lsblk
NAME        MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
nvme0n1     259:0    0  8G  0 disk
├─nvme0n1p1 259:1    0  8G  0 part /
├─nvme0n1p127 259:2    0  1M  0 part
├─nvme0n1p128 259:3    0  10M  0 part /boot/efi
└─nvme1n1   259:4    0 10G  0 disk
[ec2-user@ip-10-0-11-59 ~]$ sudo mkfs -t ext4 /dev/nvme1n1
mkfs 1.46.5 (30-Dec-2021)
Creating filesystem with 2621440 4k blocks and 655360 inodes
Filesystem UUID: 79f67bf0-0127-49e0-ba58-36344721cb87
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912, 819200, 884736, 1605632

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done

[ec2-user@ip-10-0-11-59 ~]$ sudo mount /dev/nvme1n1 /data
[ec2-user@ip-10-0-11-59 ~]$ df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        4.0M   0  4.0M   0% /dev
tmpfs           459M   0  459M   0% /dev/shm
tmpfs           184M  440K  183M   1% /run
/dev/nvme0n1p1  8.0G  1.6G   6.4G  20% /
tmpfs           459M   0  459M   0% /tmp
/dev/nvme0n1p128 10M   1.3M   8.7M  13% /boot/efi
tmpfs           92M   0   92M   0% /run/user/1000
/dev/nvme1n1    9.8G  24K   9.3G   1% /data
[ec2-user@ip-10-0-11-59 ~]$

```

5. Install Web Server and Host Sample Page

- Install Apache
 - *sudo yum update -y*
 - *sudo yum install httpd -y*
 - *sudo systemctl start httpd*
 - *sudo systemctl enable httpd*

```

Complete!
[ec2-user@ip-10-0-11-59 ~]$ sudo systemctl start httpd
[ec2-user@ip-10-0-11-59 ~]$ sudo systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[ec2-user@ip-10-0-11-59 ~]$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Tue 2026-01-13 07:57:11 UTC; 15s ago
     Docs: man:httpd.service(8).
  Main PID: 19600 (httpd)
    Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"
    Tasks: 177 (limit: 1067)
   Memory: 13.3M
      CPU: 81ms
  CGroup: /system.slice/httpd.service
          └─19600 /usr/sbin/httpd -DFOREGROUND
            └─19733 /usr/sbin/httpd -DFOREGROUND
              └─19735 /usr/sbin/httpd -DFOREGROUND
                └─19736 /usr/sbin/httpd -DFOREGROUND
                  └─19739 /usr/sbin/httpd -DFOREGROUND

Jan 13 07:57:11 ip-10-0-11-59.ap-south-1.compute.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Jan 13 07:57:11 ip-10-0-11-59.ap-south-1.compute.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Jan 13 07:57:11 ip-10-0-11-59.ap-south-1.compute.internal httpd[19600]: Server configured, listening on: port 80
[ec2-user@ip-10-0-11-59 ~]$

```

- Create Sample Web Page
 - *cd /var/www/html*
 - *Sudo nano index.html*
 - *<h1> Welcome to AWS Web Application</h1>*

Welcome to AWS Web Application

6. Create Application Load Balancer (ALB)

Before create target group

Create Target Group

- EC2 -Target Groups → Create
- Type: Instances
- Protocol: HTTP
- Port: 80
- Register EC2 instance

EC2 > Target groups > Create target group

Step 1
Create target group

Step 2 - recommended
Register targets

Step 3
Review and create

Create target group

A target group can be made up of one or more targets. Your load balancer routes requests to the targets in a target group and performs health checks on the targets.

Settings - immutable

Choose a target type and the load balancer and listener will route traffic to your target. These settings can't be modified after target group creation.

Target type

Indicate what resource type you want to target. Only the selected resource type can be registered to this target group.

☒ **Instances**
Supports load balancing to instances in a VPC. Integrate with Auto Scaling Groups or ECS services for automatic management.
Suitable for: ALB NLB GWLB

☐ **IP addresses**
Supports load balancing to VPC and on-premises resources. Facilitates routing to IP addresses and network interfaces on the same instance. Supports IPv6 targets.
Suitable for: ALB NLB GWLB

☐ **Lambda function**
Supports load balancing to a single Lambda function. ALB required as traffic source.
Suitable for: ALB

☐ **Application Load Balancer**
Allows use of static IP addresses and PrivateLink with an Application Load Balancer. NLB required as traffic source.
Suitable for: NLB

Target group name

Name must be unique per Region per AWS account.

mywebserver

Accepts: a-z, A-Z, 0-9, and hyphen (-). Can't begin or end with hyphen. 1-32 total characters; Count: 11/32

Protocol
Protocol for communication between the load balancer and targets.

Port
Port number where targets receive traffic. Can be overridden for individual

EC2 > Target groups > Create target group

mywebserver

Accepts: a-z, A-Z, 0-9, and hyphen (-). Can't begin or end with hyphen. 1-32 total characters; Count: 11/32

Protocol
Protocol for communication between the load balancer and targets.

HTTP

Port
Port number where targets receive traffic. Can be overridden for individual targets during registration.

80

1-65535

IP address type
Only targets with the indicated IP address type can be registered to this target group.

☒ **IPv4**
Each instance has a default network interface (eth0) that is assigned the primary private IPv4 address. The instance's primary private IPv4 address is the one that will be applied to the target.

☐ **IPv6**
Each instance you register must have an assigned primary IPv6 address. This is configured on the instance's default network interface (eth0). [Learn more](#)

VPC
Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.

vpc-04f2c67356d87b57a (My-VPC--vpc)

10.0.0.0/16

[Create VPC](#)

Protocol version

☒ **HTTP1**
Send requests to targets using HTTP/1.1. Supported when the request protocol is HTTP/1.1 or HTTP/2.

☐ **HTTP2**
Send requests to targets using HTTP/2. Supported when the request protocol is HTTP/2 or gRPC, but gRPC-specific features are not available.

Step 1
Create target group

Step 2 - recommended
Register targets

Step 3
Review and create

Register targets - recommended
This is an optional step to create a target group. However, to ensure that your load balancer routes traffic to this target group you must register your targets.

Available instances (1/1)

Filter instances

☒	Instance ID	Name	State	Security groups	Zone	Private IP
☒	i-005688ba2cc38c349	Myweb-server	Running	Myweb-server	ap-south-1a	10.0.11.59

1 selected

Ports for the selected instances
Ports for routing traffic to the selected instances.

80

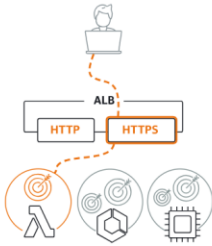
1-65535 (separate multiple ports with commas)

[Include as pending below](#)

1 selection is now pending below. Include more or register targets when ready.

- EC2 - Load Balancers – Create
- Select Application Load Balancer
 - Scheme: Internet-facing
 - Listener: HTTP 80
- Select at least 2 subnets
- Security group:
 - Allow HTTP 80
- Attach target group

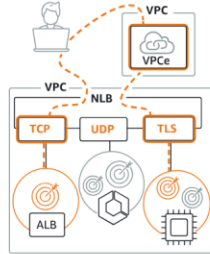
Application Load Balancer [Info](#)



Choose an Application Load Balancer when you need a flexible feature set for your applications with HTTP and HTTPS traffic. Operating at the request level, Application Load Balancers provide advanced routing and visibility features targeted at application architectures, including microservices and containers.

[Create](#)

Network Load Balancer [Info](#)



Choose a Network Load Balancer when you need ultra-high performance, TLS offloading at scale, centralized certificate deployment, support for UDP, and static IP addresses for your applications. Operating at the connection level, Network Load Balancers are capable of handling millions of requests per second securely while maintaining ultra-low latencies.

Gateway Load Balancer [Info](#)



Choose a Gateway Load Balancer when you need to deploy and manage a fleet of third-party virtual appliances that support GENEVE. These appliances enable you to improve security, compliance, and policy controls.

[Create](#)

The Application Load Balancer distributes incoming HTTP and HTTPS traffic across multiple targets such as Amazon EC2 instances, microservices, and containers, based on request attributes. When the load balancer receives a connection request, it evaluates the listener rules in priority order to determine which rule to apply, and if applicable, it selects a target from the target group for the rule action.

How Application Load Balancers work

Basic configuration

Load balancer name

Name must be unique within your AWS account and can't be changed after the load balancer is created.

Myweb-server

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme [Info](#)

Scheme can't be changed after the load balancer is created.

☒ Internet-facing

- Serves internet-facing traffic.
- Has public IP addresses.
- DNS name resolves to public IPs.
- Requires a public subnet.

☐ Internal

- Serves internal traffic.
- Has private IP addresses.
- DNS name resolves to private IPs.
- Compatible with the IPv4 and Dualstack IP address types.

Load balancer IP address type [Info](#)

Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types. Public IPv4 addresses have an additional cost.

☒ IPv4

Includes only IPv4 addresses.

☐ Dualstack

Includes IPv4 and IPv6 addresses.

☐ Dualstack without public IPv4

Network mapping [Info](#)

The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

VPC [Info](#)

The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to Lambda or on-premises targets, or if using VPC peering. To confirm the VPC for your targets, view [target groups](#).

vpc-04f2c67356d87b37a (My-VPC--vpc)

10.0.0.0/16



[Create VPC](#)

IP pools [Info](#)

You can optionally choose to configure an IPAM pool as the preferred source for your load balancers IP addresses. Create or view [Pools](#) in the [Amazon VPC IP Address Manager console](#).

☐ Use IPAM pool for public IPv4 addresses

The IPAM pool you choose will be the preferred source of public IPv4 addresses. If the pool is depleted IPv4 addresses will be assigned by AWS.

Availability Zones and subnets [Info](#)

Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.

☒ ap-south-1a (aps1-az1)

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-0276eadab9a1d009d
IPv4 subnet CIDR: 10.0.0.0/20

My-VPC--subnet-public1-ap-south-1a

☒ ap-south-1b (aps1-az3)

Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-00726d667361f2cb1
IPv4 subnet CIDR: 10.0.16.0/20

My-VPC--subnet-public2-ap-south-1b

EC2 > Load balancers > Create Application Load Balancer

Listeners and routing Info

A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

▼ Listener HTTP:80 Remove

Protocol

Port
1-65535

Default action Info

The default action is used if no other rules apply. Choose the default action for traffic on this listener.

Routing action

☒ Forward to target groups
 ☐ Redirect to URL
 ☐ Return fixed response

Forward to target group Info

Choose a target group and specify routing weight or [create target group](#).

Target group	Weight	Percent
mywebserver Target type: Instance, IPv4 Target stickiness: Off	1 0-999	100%

[+ Add target group](#)

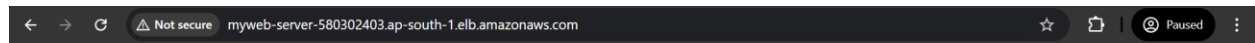
You can add up to 4 more target groups.

Target group stickiness Info

Enables the load balancer to bind a user's session to a specific target group. To use stickiness the client must support cookies. If you want to bind a user's session to a specific target, turn on the Target Group attribute Stickiness.

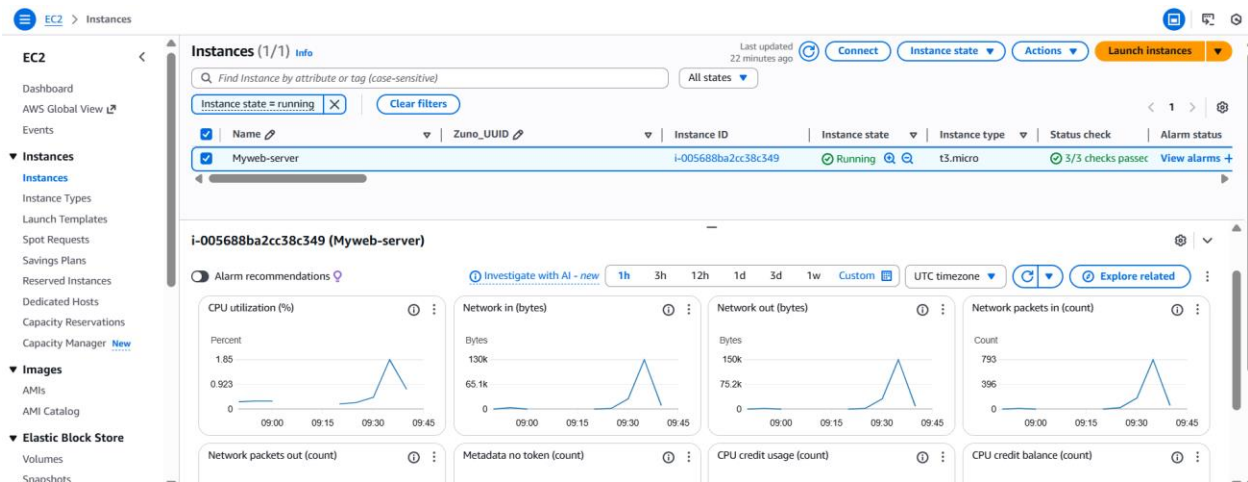
Test

<http://<<ALB-DNS>>>



7. Enable CloudWatch Monitoring

- EC2 → Select instance
- Monitoring tab → CPU Utilization visible



8. Create SNS Topic and Email Subscription

- Go to SNS - Topics
 - Create Topic:
 - Type: Standard
 - Name: CPUAlertTopic

Amazon SNS > Topics > Create topic

Create topic

Details

Type: [Info](#)
Topic type cannot be modified after topic is created

☐ FIFO (first-in, first-out)

- Strictly-preserved message ordering
- Exactly-once message delivery
- Subscription protocols: SQS

☒ Standard

- Best-effort message ordering
- At-least once message delivery
- Subscription protocols: SQS, Lambda, Data Firehose, HTTP, SMS, email, mobile application endpoints

Name

CPUAlert

Maximum 256 characters. Can include alphanumeric characters, hyphens (-) and underscores (_).

Display name - optional [Info](#)
To use this topic with SMS subscriptions, enter a display name. Only the first 10 characters are displayed in an SMS message.

My Topic

Maximum 100 characters.

► **Encryption - optional**
Amazon SNS provides in-transit encryption by default. Enabling server-side encryption adds at-rest encryption to your topic.

Amazon SNS > Topics > CPULAlert

Topic CPULAlert created successfully. You can create subscriptions and send messages to them from this topic. [Publish message](#)

Notifications: 0 0 0 1 0 0

CPULAlert [Edit](#) [Delete](#) [Publish message](#)

Details

Name CPULAlert	ARN arn:aws:sns:ap-south-1:989615686095:CPULAlert	Display name CPULAlertTopic	Type Standard
--------------------------	---	---------------------------------------	-------------------------

Topic owner
989615686095

[Subscriptions](#) [Access policy](#) [Data protection policy](#) [Delivery policy \(HTTP/S\)](#) [Delivery status logging](#) [Encryption](#) [Tags](#) [Integrations](#)

Subscriptions (0) [Edit](#) [Delete](#) [Request confirmation](#) [Confirm subscription](#) [Create subscription](#)

Search

ID	Endpoint	Status	Protocol
No subscriptions found You don't have any subscriptions to this topic.			

- Create Subscription
 - Inside topic - Create subscription
 - Protocol: Email
 - Endpoint: Your email address
- Confirm

Amazon SNS > Subscriptions > Create subscription

Create subscription

Details

Topic ARN
arn:aws:sns:ap-south-1:989615686095:CPULAlert

Protocol
The type of endpoint to subscribe
Email

Endpoint
An email address that can receive notifications from Amazon SNS.
test@example.com

After your subscription is created, you must confirm it. [Info](#)

► **Subscription filter policy - optional** [Info](#)
This policy filters the messages that a subscriber receives.

► **Redrive policy (dead-letter queue) - optional** [Info](#)

CPULAlertTopic <no-reply@sns.amazonaws.com> [👍](#) [👎](#) [🔄](#) [📧](#) [⋮](#)

To: 🕒 Ajit Jadhav Tue 1/13/2026 3:27 PM

You have chosen to subscribe to the topic:

To confirm this subscription, click or visit the link below (If this was in error no action is necessary):
[Confirm subscription](#)

Please do not reply directly to this email. If you wish to remove yourself from receiving all future SNS subscription confirmation requests please send an email to sns-opt-out

[🔄 Reply](#) [➡ Forward](#)



Simple Notification Service

Subscription confirmed!

You have successfully subscribed.

Your subscription's id is:

[REDACTED]

If it was not your intention to subscribe, [click here to unsubscribe](#).

9. Create CloudWatch Alarm (High CPU)

- CloudWatch - Alarms - Create Alarm
- Select metric:
 - EC2 -Per-Instance Metrics - *CPUUtilization*
 - Condition:
 - Threshold: Greater than 70%
 - Period: 5 minutes
- Notification:
 - Select SNS topic created
- Alarm name: *CPUALart*

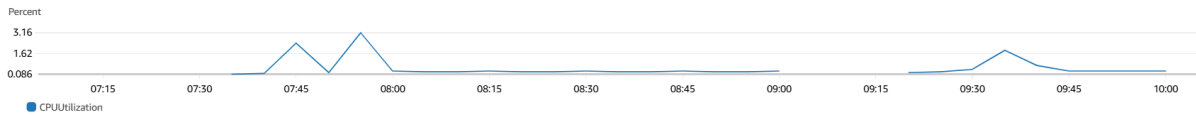
The screenshot shows the AWS CloudWatch Alarms console. The left sidebar contains navigation links for Ingestion, Dashboards, Alarms (with a sub-link for 'In alarm'), AI Operations, GenAI Observability, Application Signals (APM), Infrastructure Monitoring, and Logs. The main area is titled 'Alarms (0)' and displays a message: 'No alarms. No alarms to display. Read more about Alarms. Create alarm'. Above this message are filters for 'Alarm state: In alarm', 'Alarm type: Any', and 'Actions status: Any'. At the top right, there are buttons for 'Clear selection', 'Create composite alarm', and 'Create alarm'.

Select metric



CPUUtilization

1h 3h 12h 1d 3d 1w Custom UTC timezone Line



Browse (113)	Multi source query	Graphed metrics (1)	Options	Source	
☐	Myweb-server	i-005688ba2cc38c349	StatusCheckFailed_Instance	No alarms	
☐	Myweb-server	i-005688ba2cc38c349	StatusCheckFailed	No alarms	
☐	Myweb-server	i-005688ba2cc38c349	StatusCheckFailed_System	No alarms	
☒	Myweb-server	i-005688ba2cc38c349	CPUUtilization	No alarms	
☐	Myweb-server	i-005688ba2cc38c349	NetworkOut	No alarms	
☐	Myweb-server	i-005688ba2cc38c349	NetworkPacketsOut	No alarms	
☐	Myweb-server	i-005688ba2cc38c349	NetworkPacketsIn	No alarms	
☐	Myweb-server	i-005688ba2cc38c349	NetworkIn	No alarms	

Cancel Select metric

CloudWatch > Alarms > Create alarm

Add alarm details
Step 4
Preview and create

This alarm will trigger when the blue line goes above the red line for 1 datapoints within 5 minutes.

Percent

70

35

0.086

07:30 08:00 08:30 09:00 09:30 10:00

CPUUtilization

Namespace

AWS/EC2

Metric name

CPUUtilization

InstancedId

i-005688ba2cc38c349

Instance name

Myweb-server

Statistic

Average

Period

5 minutes

Conditions

Threshold type

☒ Static Use a value as a threshold

☐ Anomaly detection Use a band as a threshold

CloudWatch > Alarms > Create alarm

5 minutes

Conditions

Threshold type

☒ Static Use a value as a threshold

☐ Anomaly detection Use a band as a threshold

Whenever CPUUtilization is...

Define the alarm condition.

☒ Greater > threshold

☐ Greater/Equal >= threshold

☐ Lower/Equal <= threshold

☐ Lower < threshold

than...

Define the threshold value.

70

Must be a number.

Additional configuration

Cancel Next

CloudWatch > Alarms > Create alarm

Step 2: **Configure actions**

Notification

Alarm state trigger
Define the alarm state that will trigger this action.

☒ **In alarm**
The metric or expression is outside of the defined threshold.

☐ **OK**
The metric or expression is within the defined threshold.

☐ **Insufficient data**
The alarm has just started or not enough data is available.

Send a notification to the following SNS topic
Define the SNS (Simple Notification Service) topic that will receive the notification.

☒ **Select an existing SNS topic**

☐ Create new topic

☐ Use topic ARN to notify other accounts

Send a notification to...

Q CPUAlert

Only topics belonging to this account are listed here. All persons and applications subscribed to the selected topic will receive notifications.

Email (endpoints) - View in SNS Console

Add notification

Lambda action

CloudWatch > Alarms > Create alarm

Alarm recommendations available
Turn on Recommendations to pre-populate the wizard with the recommended alarms.

Step 3: **Add alarm details**

Name and description

Alarm name
CPUAlert

Alarm description - optional View formatting guidelines

Edit Preview

This is an H1
double asterisks will produce strong character
This is [an example](https://example.com/) inline link.

Up to 1024 characters (0/1024)

Markdown formatting is only applied when viewing your alarm in the console. The description will remain in plain text in the alarm notifications.

CloudWatch > Alarms

Alarms (1)

Hide Auto Scaling alarms

Clear selection Create composite alarm Actions Create alarm

Search Alarm state: Any Alarm type: Any Actions status: Any

	Name	State	Last state update (UTC)	Conditions	Actions
☐	CPUAlert	OK	2026-01-13 10:11:54	CPUUtilization > 70 for 1 datapoints within 5 minutes	Actions enabled

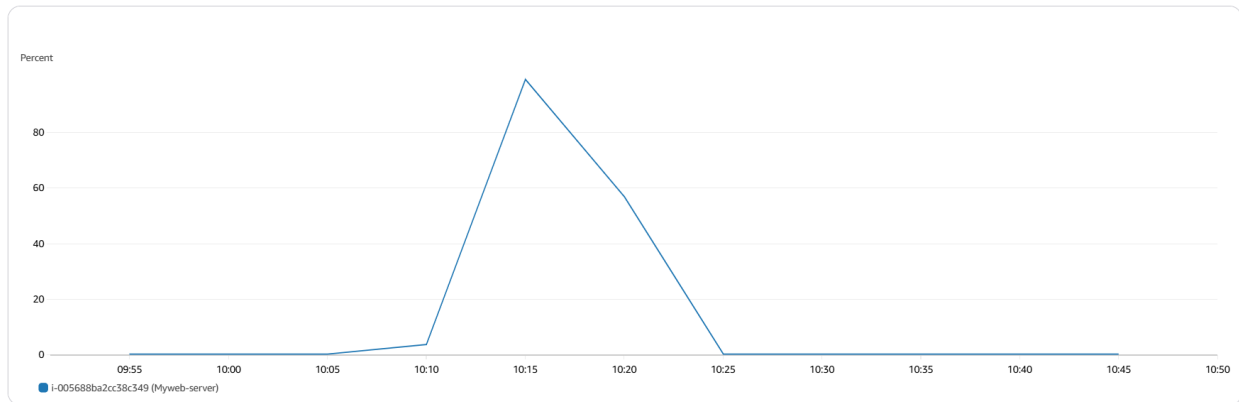
Test:

```
sudo yum install stress -y
```

```
stress --cpu 2 --timeout 300
```


CPU utilization (%)

1 minute Average 1h 3h 12h 1d 3d 1w Custom UTC timezone



View in metrics Close

Received Notification :

ALARM: "CPUALert" in Asia Pacific (Mumbai)

Summarize

Threshold:

- The alarm is in the ALARM state when the metric is GreaterThanThreshold 70.0 for at least 1 of the last 1 period(s) of 300 seconds.

Monitored Metric:

- MetricNamespace: AWS/EC2
- MetricName: CPUUtilization
- Dimensions: [InstanceId = i-005688ba2cc38c349]
- Period: 300 seconds
- Statistic: Average
- Unit: not specified
- TreatMissingData: missing

State Change Actions:

- OK:
- ALARM: [arn:aws:sns:ap-south-:CPUALert]
- INSUFFICIENT_DATA:

--

If you wish to stop receiving notifications from this topic, please click or visit the link below to unsubscribe:

Please do not reply directly to this email. If you have any questions or comments regarding this email, please contact us at <https://aws.amazon.com/support>

